Example 5.21 If the centroid of a triangle is at $(-2,1)$ and two of its vertices are $(1,-6)$ and $(-5,2)$, then find the third vertex of the triangle.

Solution Let the vertices of a triangle be

$A(1,-6)$, $B(-5,2)$ and $C(x_3, y_3)$

Given the centroid of a triangle as $(-2,1)$ we get,

$$\frac{x_1 + x_2 + x_3}{3} = -2 \qquad \frac{y_1 + y_2 + y_3}{3} = 1$$

$$\frac{1 - 5 + x_3}{3} = -2 \qquad \frac{-6 + 2 + y_3}{3} = 1$$

$$-4 + x_3 = -6 \qquad -4 + y_3 = 3$$

$$x_3 = -2 \qquad y_3 = 7$$

Therefore, third vertex is $(-2,7)$.

Thinking Corner

(i) Master gave a triangular plate with vertices A(5, 8), B(2, 4), C(8, 3) and a stick to a student. He wants to balance the plate on the stick. Can you help the boy to locate that point which can balance the plate.

(ii) Which is the centre of gravity for this triangle? why?

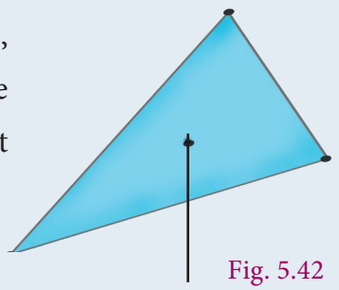


Fig. 5.42

Exercise 5.5

1. Find the centroid of the triangle whose vertices are

   (i) (2,–4), (–3,–7) and (7,2) (ii) (–5,–5), (1,–4) and (–4,–2)

2. If the centroid of a triangle is at (4,–2) and two of its vertices are (3,–2) and (5,2) then find the third vertex of the triangle.

3. Find the length of median through A of a triangle whose vertices are $A(-1,3)$, $B(1,-1)$ and $C(5,1)$.

4. The vertices of a triangle are (1,2), $(h,-3)$ and $(-4,k)$. If the centroid of the triangle is at the point (5,–1) then find the value of $\sqrt{(h+k)^2 + (h+3k)^2}$.

5. Orthocentre and centroid of a triangle are $A(-3,5)$ and $B(3,3)$ respectively. If C is the circumcentre and AC is the diameter of this circle, then find the radius of the circle.

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